BEAMLINE

BUILD DOCUMENT · 25 OCTOBER 2026 DESIGNED IN BEAMLINE

A build manual for a tabletop laser-optics setup

aom doublepass fixed

Overview

Rb-87 780 nm AOM double-pass (fixed: return ghost dumped) in cat-eye configuration (after PyOpticL's doublepass module). Fiber-coupled light is cleaned up by a HWP + PBS, diffracted by an 80 MHz Isomet AOM, retro-reflected through a f=75 mm lens / QWP / mirror cat-eye so the +1 order re-enters the AOM and is diffracted again (+160 MHz total, beam pointing independent of RF frequency). The double-passed QWP turns H into V so the PBS reflects the return beam into the output fiber. The forward 0th order is caught by a razor-blade dump before the retro mirror, and the un-diffracted part of the return pass (a ghost 0.85° off axis) is blocked before the output fiber.

Bragg angle at 80 MHz, 780 nm, TeO2 (4200 m/s): 14.9 mrad. Cat-eye: AOM–lens = lens–mirror = f, so the return beam retraces the +1 path for any RF frequency. Forward 0th order is offset by f·θ ≈ 1.1 mm at the retro mirror and is blocked there.

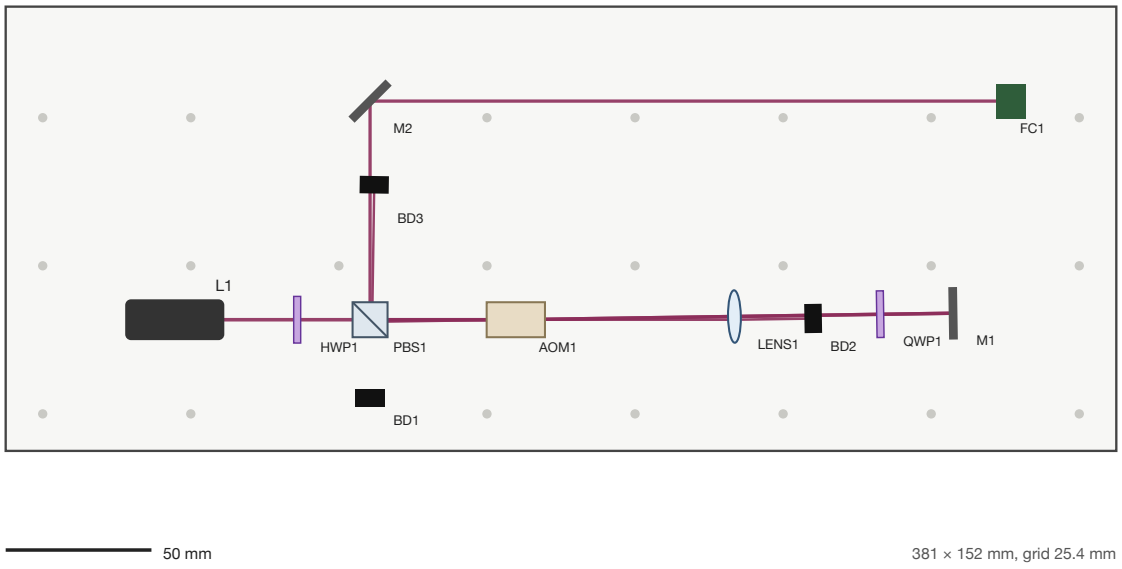


Figure 1: Plan view. Beams coloured by wavelength (infrared shown dark magenta), width ∝ log power; dashed = leaves the plate; grey dots = table holes.

Key specifications

Quantity	Value
Wavelengths	780.24 nm
Input power	20 mW from 1 source(s)
Delivered (fibers + detectors)	11.8 mW
Elements / beam segments	11 / 16
Baseplate	381 × 152.4 × 25.4 mm, 2 printed tile(s)
Fiber FC1	11.8 mW, coupling 92 %

Verification

PASS score 98 / 100 (Beamline tracer: 2D ray + Gaussian beam + Jones polarization)

Targets

Target	Status
met: FC1: 11.796 mW (need ≥ 8.000)	met
met: FC1: coupling 92.2% (need ≥ 80.0%)	met
met: FC1: polarization V (need V, overlap 100.0%)	met

Checks

Severity	Check	Message
INFO	target	met: FC1: 11.796 mW (need ≥ 8.000)
INFO	target	met: FC1: coupling 92.2% (need ≥ 80.0%)
INFO	target	met: FC1: polarization V (need V, overlap 100.0%)

Bill of materials

Purchased parts

Part	Description	Vendor	Qty	Unit	Line
AC127-075-B	lens (Cat-eye lens f=75 mm)	Thorlabs	1	$90.00	$90.00
BB05-E03	mirror (Cat-eye retro mirror (focal plane))	Thorlabs	2	$65.00	$130.00
KM100	kinematic mirror mount, Ø1"	Thorlabs	2	$48.00	$96.00
KM100CP	kinematic mount for 10 mm cubes	Thorlabs	1	$70.00	$70.00
LB1	beam block / dump, black anodized	Thorlabs	3	$95.00	$285.00
LMR1	lens mount, Ø1" optics	Thorlabs	1	$18.00	$18.00
PAF2-A4A	Fiber input (PM780-HP via PAF2-A4A collimator) (780.24 nm, 20 mW)	Thorlabs	2	$420.00	$840.00
PBS052	10 mm polarizing beamsplitter cube, 620-1000 nm	Thorlabs	1	$260.00	$260.00
RSP05	rotation mount, Ø1/2" optics	Thorlabs	2	$95.00	$190.00
WPH05M-780	half-wave plate (Input λ/2 (rotate to H))	Thorlabs	1	$310.00	$310.00
WPQ05M-780	quarter-wave plate (λ/4 at 45° (double pass → λ/2))	Thorlabs	1	$290.00	$290.00
Isomet 1205C-2	acousto-optic modulator (80 MHz AOM)	Isomet	1	$1,500.00	$1,500.00

Estimated total: $4,095.00

list-price estimates, USD, excluding lasers; verify with vendors

Hardware

Part	Description	Qty	Unit
SH25S050	1/4-20 x 1/2" cap screw (25 pack -> per piece)	22	$0.40
SH8S100	8-32 x 1" cap screw (per piece)	11	$0.30
DOWEL-2x8	Ø2 x 8 mm steel dowel pin (per piece)	22	$0.15
M3x25+NUT	M3 x 25 mm cap screw + nut (tile lap joints)	3	$0.20

3D-printed parts

Part	STL file	cm³	Print h	PLA g
baseplate_r1c1	stl/baseplate_r1c1.stl	732.3	19.5	290
baseplate_r1c2	stl/baseplate_r1c2.stl	690	18.6	276

Assembly

Step 1 **Print the baseplate** — Print 2 file(s) from stl/ flat, top side up: PLA, 20 % infill, 3 walls (≈ 38.1 h, ≈ 566 g).

Step 2 **Join the tiles** — Lay the tiles out by their r/c names, overlap the 20 mm half-lap joints and fasten with 3 M3 × 25 bolts + nuts.

Step 3 **Press in the dowel pins** — Press 22 Ø2 × 8 mm pins into the paired pin holes; they key each mount's orientation.

Step 4 **Bolt to the table** — Fix the plate with 22 1/4-20 (or M6) cap screws through the counterbored holes (2-grid pitch, half a grid in from the edges).

Step 5 **HWP1** — Mount Input λ/2 (RSP05, WPH05M-780) at site 1, (100.0, 45.0) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Set the fast axis to 10.0° from horizontal (H).

Step 6 **PBS1** — Mount Polarizing beamsplitter cube (KM100CP, PBS052) at site 2, (125.0, 45.0) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Outputs: L1.t → 0.0° (20 mW); L1.t.+1.0.r → 89.1° (3.2 mW); L1.t.+1.+1.r → 90.0° (12.8 mW)

Step 7 **AOM1** — Mount 80 MHz AOM (Isomet 1205C-2) at site 3, (175.0, 45.0) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Drive at 80 MHz; use the +1 order (≈ 80 % efficiency).

Step 8 **LENS1** — Mount Cat-eye lens f=75 mm (AC127-075-B, LMR1) at site 4, (250.0, 46.1) mm; orientation 0.9°; keyed by pins, one 8-32 screw from below. f = 75 mm, 75.0 mm after the previous element on L1.t.+1.

Step 9 **BD2** — Mount 0th-order razor block (LB1) at site 5, (275.0, 45.4) mm; orientation 0.9°; keyed by pins, one 8-32 screw from below. Blocks: L1.t.0 (4 mW, 780.24 nm)

Step 10 **QWP1** — Mount λ/4 at 45° (RSP05, WPQ05M-780) at site 6, (300.0, 46.9) mm; orientation 0.9°; keyed by pins, one 8-32 screw from below. Set the fast axis to 45.0° from horizontal (H).

Step 11 **M1** — Mount Cat-eye retro mirror (BB05-E03, KM100) at site 7, (325.0, 47.2) mm; orientation 180.9°; keyed by pins, one 8-32 screw from below. Steer the beam to: L1.t.+1 → 180.9° (16 mW)

Step 12 **BD3** — Mount Return 0th-order ghost block (LB1) at site 8, (126.4, 89.3) mm; orientation 89.1°; keyed by pins, one 8-32 screw from below. Blocks: L1.t.+1.0.r (3.2 mW, 780.24 nm)

Step 13 **M2** — Mount Output fold mirror (BB05-E03, KM100) at site 9, (125.0, 120.0) mm; orientation 315.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: L1.t.+1.+1.r → 360.0° (12.8 mW)

Step 14 **FC1** — Mount Output fiber coupler (PAF2-A4A) at site 10, (340.0, 120.0) mm; orientation 360.0°; keyed by pins, one 8-32 screw from below. Collimation lens f = 4.6 mm, fiber MFD 5.3 µm. Walk the two last mirrors and adjust the focus for maximum: expect ≈ 11.8 mW (92 % coupling).

Step 15 **BD1** — Mount Dump for rejected V light (LB1) at site 11, (125.0, 20.0) mm; orientation 270.0°; keyed by pins, one 8-32 screw from below. Blocks: (nothing in this state)

Alignment procedure

Align in beam order with an iris and an IR card; adjust one knob at a time and keep the best reading.

Order: HWP1 → AOM1 → LENS1 → QWP1 → M1 → M2 → FC1

No tuner run attached: use the design values listed in the assembly steps.

Expected readings

- expect ≈ 11.8 mW at FC1, coupling 92 %

Laser safety

Wavelengths: 780.24 nm. **Maximum source power on the table:** 20 mW.

Wavelength	Band	Eyewear (generic guidance)
780 nm	near-IR (invisible!)	OD ≥ 3 at 780 nm (generic; confirm with your laser safety officer)

Beam dumps

Dump	Blocks
BD2	L1.t.0 (4 mW, 780.24 nm)
BD3	L1.t.+1.0.r (3.2 mW, 780.24 nm)
BD1	(nothing in the default state)

Verify all beam dumps before full power. Align at the lowest usable power, never look along the beam, remove watches and rings, keep the beam below eye level, and enclose invisible (IR) beams.

Credits

- Layout concept and “paint by numbers” baseplates: PyOpticL (code-to-CAD optics).
- Designed by Gemini `gemini-4-argon` agents, verified by the Beamline tracer.
- Prices are list-price estimates; part numbers should be checked against current vendor catalogs.